

Assignment No 29

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Domain : AI &ML

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Q1.

```
import pandas as pd
from sklearn.model_selection import train_test_split

df = pd.read_csv("cleaned_emotions.csv")

X = df["text"]
y = df["emotions"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
print("X_train shape:", X_train.shape)
print("X_test shape:", X_test.shape)
print("y_train shape:", y_train.shape)
print("y_test shape:", y_test.shape)
```

```
... X_train shape: (24,)
    X_test shape: (6,)
    y_train shape: (24,)
    y_test shape: (6,)
```

Q2.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer

df = pd.read_csv("cleaned_emotions.csv")

X = df["text"]
y = df["emotions"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

vectorizer = CountVectorizer()

X_train_bow = vectorizer.fit_transform(X_train)

X_test_bow = vectorizer.transform(X_test)

print("X_train BOW shape:", X_train_bow.shape)
print("X_test BOW shape:", X_test_bow.shape)

print("\nFirst 20 feature names:")
print(vectorizer.get_feature_names_out()[:20])
```

```
... X_train BOW shape: (24, 71)
    X_test BOW shape: (6, 71)
```

```
First 20 feature names:
['about' 'afraid' 'after' 'am' 'and' 'angry' 'annoyed' 'before' 'best'
 'by' 'cares' 'cheerful' 'completely' 'dark' 'day' 'decision' 'deeply'
 'delay' 'disappointed' 'down']
```

Q3.

```
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score

model = MultinomialNB()

model.fit(X_train_bow, y_train)

y_pred = model.predict(X_test_bow)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Accuracy (%)ate", accuracy * 100)

... Accuracy: 0.5
   Accuracy (%): 50.0
```

```
print("\nBow vector:")
print(sample_bow.toarray())
```

[illegible]

Q5.

```
*** X_train N-gram shape: (24, 160)
    X_test N-gram shape: (6, 160)
```

First 20 N-gram features:

```
['about' 'about everyone' 'about the' 'about this' 'about tomorrow'
 'afraid' 'afraid of' 'after' 'after the' 'am' 'am afraid' 'am annoyed'
 'am excited' 'am feeling' 'am so' 'am surprised' 'am worried' 'and'
 'and sad' 'angry']
```

Some bigram features:

```
['about everyone', 'about the', 'about this', 'about tomorrow', 'afraid of', 'after the', 'am afraid', 'am annoyed', 'am excited', 'am feeling', 'am so'
```

Q6.

```
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score

nb_ngram = MultinomialNB()

nb_ngram.fit(X_train_ngram, y_train)

y_pred_ngram = nb_ngram.predict(X_test_ngram)

ngram_accuracy = accuracy_score(y_test, y_pred_ngram)

print("Unigram (Basic BoW) Accuracy:", accuracy)
print("N-gram (Unigram + Bigram) Accuracy:", ngram_accuracy)

print("\nUnigram Accuracy (%):", accuracy * 100)
print("N-gram Accuracy (%):", ngram_accuracy * 100)

if ngram_accuracy > accuracy:
    print("\nN-gram model performed better.")
elif ngram_accuracy < accuracy:
    print("\nUnigram model performed better.")
else:
    print("\nBoth models have the same accuracy.")
```

```
... Unigram (Basic BoW) Accuracy: 0.5
    N-gram (Unigram + Bigram) Accuracy: 0.5
```

```
Unigram Accuracy (%): 50.0
N-gram Accuracy (%): 50.0
```

```
Both models have the same accuracy.
```

Q7.

```
▶ from sklearn.feature_extraction.text import TfidfVectorizer
```

```
tfidf = TfidfVectorizer()
```

```
X_train_tfidf = tfidf.fit_transform(X_train)
```

```
X_test_tfidf = tfidf.transform(X_test)
```

```
print("X_train TF-IDF shape:", X_train_tfidf.shape)
```

```
print("X_test TF-IDF shape:", X_test_tfidf.shape)
```

```
print("\nFirst 15 TF-IDF feature names:")
```

```
print(tfidf.get_feature_names_out()[:15])
```

```
... X_train TF-IDF shape: (24, 71)
```

```
X_test TF-IDF shape: (6, 71)
```

```
First 15 TF-IDF feature names:
```

```
['about' 'afraid' 'after' 'am' 'and' 'angry' 'annoyed' 'before' 'best'  
 'by' 'cares' 'cheerful' 'completely' 'dark' 'day']
```

Q8.

```
▶ from sklearn.naive_bayes import MultinomialNB
  from sklearn.metrics import accuracy_score
  |
  nb_tfidf = MultinomialNB()

  nb_tfidf.fit(X_train_tfidf, y_train)

  y_pred_tfidf = nb_tfidf.predict(X_test_tfidf)

  tfidf_accuracy = accuracy_score(y_test, y_pred_tfidf)

  print("TF-IDF + MultinomialNB Accuracy:", tfidf_accuracy)
  print("Accuracy (%)ate:", tfidf_accuracy * 100)
```

```
... TF-IDF + MultinomialNB Accuracy: 0.0
    Accuracy (%): 0.0
```

Q9.

```
import pandas as pd
|
comparison = pd.DataFrame({
    "Vectorizer": [
        "Bag of Words (Unigrams)",
        "Bag of Words (Unigrams + Bigrams)",
        "TF-IDF"
    ],
    "Accuracy": [
        accuracy,
        ngram_accuracy,
        tfidf_accuracy
    ]
})

comparison["Accuracy (%)"] = comparison["Accuracy"] * 100

print(comparison)
```

```
...      Vectorizer  Accuracy  Accuracy (%)
0      Bag of Words (Unigrams)      0.5      50.0
1  Bag of Words (Unigrams + Bigrams)      0.5      50.0
2              TF-IDF      0.0      0.0
```

Q10.

```
*** Dataset Shape: (30, 2)
```

```
First 5 rows:
```

```

              text emotions
0      i feel really happy today    joy
1      i am so angry about this    anger
2      i am worried about tomorrow    fear
3      this makes me feel sad    sadness
4 i love spending time with my family    love
```

```
Training samples: 24
```

```
Testing samples: 6
```

```
CountVectorizer Accuracy: 0.5
```

```
CountVectorizer Accuracy (%): 50.0
```

```
TF-IDF Accuracy: 0.0
```

```
TF-IDF Accuracy (%): 0.0
```

```
=====
MODEL COMPARISON
```

```
=====
CountVectorizer: 50.0 %
```

```
TF-IDF: 0.0 %
```

```
Best Vectorizer: CountVectorizer
```

```
Best Accuracy: 50.0 %
```

```
Files saved successfully!
```